Practical 5: Analyzing Sales Performance by Region in a Retail Company

Aim: To analyze retail sales data and compare sales performance across different regions using data aggregation and visualization.

Cell 1: Importing Libraries

This cell imports the required Python libraries. Pandas is used for handling and analyzing sales data, NumPy is used for numerical operations, and Matplotlib is used to create graphs. The figure size setting defines the default graph size, while the grid setting displays grid lines to make graphs easier to understand.

Important viva questions:

1. What is Pandas? — A Python library used for data manipulation and analysis.
2. What is NumPy? — A library used for numerical calculations.
3. What is Matplotlib? — A library used for data visualization.
4. Why do we use plt.rcParams? — To set default graph properties.

   CELL2-

   Cell 2: Loading the Sales Dataset

   Aim

   To load the retail sales dataset from a CSV file into a Pandas DataFrame.

   Explanation

   This cell loads the file Retail_Sales_Data.csv using pd.read_csv(). The loaded data is stored in a DataFrame named df. The try-except block handles the situation where the CSV file is not found. If the file is available, a success message is displayed. Finally, df.head() displays the first five rows of the dataset so we can check whether the data has loaded correctly.

   Cell 3: Understanding and Exploring the Dataset

   Aim

   To understand the structure, columns, data types, missing values and summary statistics of the sales dataset.

   Explanation

   This cell performs basic data exploration. First, it displays the shape of the dataset, which tells us the number of rows and columns. Then it displays all column names and their data types. Next, it checks for missing values in each column. Finally, describe(include="all") provides summary statistics such as count, mean, standard deviation, minimum and maximum values for numerical data, along with information about categorical data.

   Output from your PDF

   1. Dataset shape

   Shape: (300, 5)

   This means the dataset contains 300 rows and 5 columns.

2. Column names

['Transaction_Date', 'Region', 'Product_Category',

'Quantity', 'Sales_Amount']

3. Data types

Column	**Data type**
Transaction_Date	object
Region	object
Product_Category	object
Quantity	int64
Sales_Amount	float64

4. Missing values

All five columns have 0 missing values. This means the dataset is complete.

5. Summary statistics

Important values shown in your PDF:

Column	**Result**
Total records	300
Unique regions	4
Most frequent region	West (85 times)
Unique product categories	4
Most frequent category	Grocery (85 times)
Average quantity	5.02
Average sales amount	2536.28
Standard deviation of sales	1461.81

SHRUTIKA CL1 CL2 (1)(1).pdf

Important Viva Questions

1. What is df.shape?
It returns the number of rows and columns.

2. What is df.columns?
It displays all column names.

3. What is df.dtypes?
It displays the data type of each column.

4. What is isna().sum()?
It counts missing values in each column.

5. What is describe()?
It provides summary statistics of the dataset.

6. What does mean() represent?
Average value.

7. What does std represent?
Standard deviation, which measures how spread out the values are.

Cell 4: Identifying Relevant Columns

Aim: To automatically identify the important columns required for data aggregation.

Explanation:

This cell defines a function called find_col() that searches the dataset and identifies columns based on their names.

- region_col identifies the Region column.

- sales_col identifies the Sales_Amount column.

- category_col identifies the Product_Category column.

- pd.to_numeric() converts the sales column into numeric values. If any value cannot be converted, it replaces it with NaN.

The main purpose is to identify the required columns automatically so we can use them in the next steps.

Viva Questions

1. What is the purpose of find_col()?
   To find a column based on its name.
2. What is pd.to_numeric()?
   It converts values into numeric data.
3. What does errors="coerce" do?
   It converts invalid values into NaN.

Cell 5: Total Sales by Region

Aim: To calculate and compare the total sales of each region.

Explanation:

This cell uses groupby() to group the dataset according to regions and calculates the total sales of each region using sum(). Then, sort_values(ascending=False) arranges the regions from highest sales to lowest sales.

It displays the total sales in a table and creates a bar graph to compare sales across different regions.

Graph explanation:

- X-axis: Different regions.

- Y-axis: Total sales amount.
- Bars: Each bar represents the total sales of one region.
- Taller bar means higher sales.

Viva Questions

1. What is groupby()?
   It groups data based on a particular column.
2. What is the use of sum()?
   It calculates the total sales.
3. What is a bar graph?
   A graph used to compare values between different categories.
4. What does ascending=False mean?
   It sorts values from highest to lowest.
5. What is plt.show()?
   It displays the graph.

Cell 6: Sales Distribution by Region

Aim: To show the percentage contribution of each region to total sales.

Explanation:

This cell again groups the data by region and calculates the total sales using groupby() and sum(). It then creates a pie chart to show how much each region contributes to the overall sales.

The autopct="%1.1f%%" displays the percentage on each slice, and startangle=90 sets the starting angle of the pie chart.

Graph explanation:

- Each slice represents one region.
- The size of each slice represents that region's total sales.
- The percentage shows that region's share of total sales.
- A larger slice means a higher contribution to sales.

Viva Questions

1. What is a pie chart?
   A circular graph used to show the percentage distribution of data.
2. What is autopct?
   It displays percentages on the pie chart.
3. What is startangle=90?
   It sets the starting angle of the pie chart to 90 degrees.

4. What does groupby() do here?
   It groups sales according to region.

5. What is the difference between a bar graph and a pie chart?
   A bar graph compares total sales, while a pie chart shows the percentage contribution of each region.

   Remember these three cells:

- Cell 4: Identifies the required columns.
- Cell 5: Calculates and displays total sales by region using a bar graph.
- Cell 6: Displays the percentage distribution of sales using a pie chart.

Cell 7: Identifying Top-Performing Regions

Aim: To find the regions that generate the highest sales.

Explanation:

This cell groups the data according to region and calculates the total sales of each region. It sorts the sales in descending order, meaning the highest sales appear first.

- head() selects the top five regions.
- display() displays the top-performing regions with their total sales.
- region_sales.index[0] gets the name of the region with the highest sales.
- len(region_sales) checks whether the result contains any regions.

Output meaning:

It displays:

1. The top five regions based on total sales.
2. The name of the region having the highest sales.

Viva Questions

1. What is head()?
It displays the first five records by default.

2. What is index[0]?
It accesses the first index value, which here is the highest-sales region.

3. What is a top-performing region?
A region that generates higher total sales compared to other regions.

4. Why do we use ascending=False?
To arrange sales from highest to lowest.

Cell 8: Sales by Region and Product Category

Aim: To calculate sales for each product category within every region.

Explanation:

This cell groups the dataset using two columns: Region and Product_Category. It calculates the total sales for every region-category combination.

- groupby([region_col, category_col]) groups data by region and product category.
- sum() calculates the total sales.
- unstack(fill_value=0) converts the product categories into separate columns.
- fill_value=0 replaces missing combinations with zero.
- display() displays the final table.

Output meaning:

The output is a table where:

- Rows represent different regions.
- Columns represent product categories.
- Values represent the total sales of each product category in each region.

Viva Questions

1. What is groupby()?
It groups data based on one or more columns.

2. Why do we use two columns in groupby()?
To calculate sales based on both region and product category.

3. What is unstack()?
It converts one level of grouped data into columns.

4. What is fill_value=0?
It replaces missing values with zero.

5. What is data aggregation?
Data aggregation is the process of grouping data and calculating summary values such as sum, average, or count.

Cell 9: Stacked Bar Graph – Sales by Region and Product Category

Aim: To compare total sales of different product categories in each region using a stacked bar graph.

Explanation:

This cell uses the region_category_sales table created in the previous cell and generates a stacked bar graph.

- kind="bar" creates a bar graph.
- stacked=True places the product category bars on top of each other.
- figsize=(10, 5) sets the graph size.
- plt.title() gives the graph a title.

- plt.xlabel() and plt.ylabel() name the X-axis and Y-axis.
- plt.legend() displays the product categories.
- plt.tight_layout() adjusts the graph layout.
- plt.show() displays the graph.

Graph explanation:

- X-axis: Different regions.
- Y-axis: Total sales amount.
- Different colours: Different product categories.
- Complete bar height: Total sales of all product categories in that region.

A taller stacked bar indicates higher overall sales in that region.

Viva Questions

1. What is a stacked bar graph?
It is a bar graph in which different categories are stacked on top of each other.

2. What is stacked=True?
It displays the categories on top of one another.

3. What does figsize=(10,5) mean?
It sets the width to 10 inches and height to 5 inches.

4. Why do we use a legend?
To identify the different product categories.

Cell 10: Grouped Bar Graph – Sales Comparison by Region and Product Category

Aim: To compare the sales of different product categories separately within each region.

Explanation:

This cell uses the same region_category_sales table but creates a grouped bar graph instead of a stacked bar graph.

- kind="bar" creates a bar graph.
- stacked=False displays the bars side by side.
- Each region contains separate bars for different product categories.
- The remaining commands add the title, axis labels, legend and display the graph.

Graph explanation:

- X-axis: Different regions.
- Y-axis: Total sales amount.
- Different colours: Different product categories.
- Bars side by side: Help compare category sales within the same region.

Viva Questions

1. What is a grouped bar graph?
It displays multiple bars side by side to compare different categories.

2. What is stacked=False?
It displays bars separately instead of placing them on top of each other.

3. What is the difference between stacked and grouped bar graphs?
A stacked graph places bars on top of each other, whereas a grouped graph places bars side by side.

4. Why do we use grouped bar graphs?
To compare different product categories across regions.

Practical 5 – Quick Revision

Cell	What it does
4	Identifies relevant columns
5	Calculates total sales by region and creates a bar graph
6	Shows sales distribution using a pie chart
7	Identifies top-performing regions
8	Calculates sales by region and product category
9	Creates a stacked bar graph
10	Creates a grouped bar graph

In this practical, we analyzed the sales performance of a retail company using Python in Jupyter Notebook. First, we imported the required libraries such as Pandas, NumPy and Matplotlib, and loaded the retail sales dataset containing transaction date, region, product category, quantity and sales amount. Then, we explored the dataset by checking its number of rows and columns, data types, missing values and summary statistics. After that, we identified the important columns, such as Region, Sales_Amount and Product_Category. We grouped the sales data according to regions and calculated the total sales for each region. We created a bar graph to compare regional sales and a pie chart to show the percentage contribution of each region. Next, we identified the top-performing regions and calculated sales for different product categories in each region. Finally, we created a stacked bar graph and a grouped bar graph to compare product category sales across different regions. Overall, this practical helped us understand data aggregation and visualize sales performance using Python.